

PM Case Study: Delivery ETA Intelligence Platform for E-commerce Logistics

Executive Summary

Customers purchasing products online often experience uncertainty once an order reaches the "Out for Delivery" stage. Current e-commerce platforms typically provide broad delivery windows such as "Arriving Today" or "Delivery by 11 PM," forcing customers to remain available throughout the day, especially for open-box deliveries and high-value purchases.

This case study proposes an ETA Intelligence Platform that sits on top of existing logistics infrastructure and converts operational delivery data into customer-facing delivery predictions. The goal is to reduce customer uncertainty, improve delivery success rates, and decrease support costs without modifying the underlying logistics network.

Problem Statement

Current Experience

When an order reaches the nearest delivery hub and is marked "Out for Delivery," customers receive limited visibility:

- Out for Delivery
- Arriving Today
- Delivery by End of Day

This creates several challenges:

Customer Problems

- Customers stay home unnecessarily.
- Open-box deliveries require physical presence.
- Customers cannot plan meetings or travel.
- High anxiety around delivery timing.
- Repeated app refreshes and support inquiries.

Business Problems

- Increased WISMO (Where Is My Order) support tickets.
 - Failed delivery attempts.
 - Additional re-delivery costs.
 - Lower customer satisfaction scores.
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Root Cause Analysis

The issue is not a lack of logistics intelligence.

Most logistics providers already possess:

- Route optimization systems
- Rider assignment systems
- GPS tracking
- Parcel scanning
- Delivery completion tracking
- Internal dashboards

The problem is that operational visibility remains internal and is not translated into customer-friendly delivery predictions.

Opportunity

Convert existing logistics signals into actionable customer-facing delivery windows.

Current Model:

Internal Logistics Data → Operations Team

Proposed Model:

Internal Logistics Data → Prediction Engine → Customer Delivery Window

Proposed Solution

ETA Intelligence Platform

A software layer that integrates with existing logistics systems and predicts delivery windows using:

Inputs

- Route assignment
- Rider allocation
- Package sequence
- GPS location
- Traffic conditions
- Historical route data
- Delivery completion events
- Failed delivery patterns

Outputs

- Dynamic delivery windows
 - Confidence scores
 - Customer notifications
 - Progressive ETA updates
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Customer Experience

Existing Experience

8:00 AM

Out for Delivery

Delivery by 11:00 PM

Customer uncertainty: Very High

Proposed Experience

8:00 AM

Expected Delivery:
2 PM – 6 PM

Confidence:
78%

12:00 PM

Expected Delivery:
2 PM – 4 PM

Confidence:
85%

2:00 PM

Expected Delivery:
2:30 PM – 3:15 PM

Confidence:
92%

Customer uncertainty: Low

Product Strategy

Phase 1 – Smart Delivery Slots

Provide broad delivery windows after route assignment.

Examples:

- Morning (9 AM – 12 PM)
- Afternoon (12 PM – 3 PM)
- Evening (3 PM – 6 PM)
- Late Evening (6 PM – 9 PM)

Goal:
Reduce customer uncertainty.

Phase 2 – Dynamic ETA Updates

Continuously update delivery windows based on:

- Completed deliveries
- Traffic conditions
- Route progress

Goal:
Improve ETA accuracy.

Phase 3 – Customer Notifications

Examples:

"Your delivery is expected between 2 PM and 4 PM."

"Your delivery is approximately 45 minutes away."

Goal:
Increase customer preparedness.

Phase 4 – Delivery Intelligence Dashboard

For logistics partners:

- ETA accuracy metrics
- Failed delivery analysis
- Route efficiency insights
- Customer engagement analytics

Goal:
Create enterprise value.

Target Customers

Primary Market

D2C Brands

Pain Point:

High WISMO support volume.

Examples:

- Fashion brands
- Electronics brands
- Beauty brands

Regional Logistics Providers

Pain Point:

Lack of advanced customer tracking systems.

Hyperlocal Delivery Networks

Pain Point:

Customer visibility and communication.

Secondary Market

Large logistics operators:

- Amazon Logistics
- Ekart
- Delhivery
- Shadowfax
- XpressBees

These customers are likely enterprise opportunities after validation.

Business Model

SaaS + Usage Pricing

Base Subscription

₹20,000/month

Includes:

- Dashboard
 - Analytics
 - ETA engine
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Usage Pricing

₹0.20 per shipment

Example:

1,000,000 shipments/month

Revenue:

₹200,000/month

Financial Analysis

Initial Investment

Team:

- 1 Product Manager
- 2 Backend Engineers
- 1 Frontend Engineer

- 1 Data Engineer
- 1 QA Engineer

Estimated Cost:

₹1–1.2 Crore (Year 1)

Includes:

- Salaries
 - Cloud infrastructure
 - Maps APIs
 - Monitoring tools
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Revenue Projection

Year 1

10 Customers

50,000 Shipments per Customer per Month

Total Shipments:

500,000/month

Revenue:

₹100,000/month

Annual Revenue:

₹12 Lakh

Objective:

Validation

Year 3

100 Customers

100,000 Shipments per Customer per Month

Total Shipments:

10 Million/month

Revenue:

₹20 Lakh/month

Annual Revenue:

₹2.4 Crore

Objective:

Scale

Key Success Metrics

Customer Metrics

- Delivery Window Accuracy
 - Customer Satisfaction Score
 - App Engagement Rate
 - Notification Open Rate
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Business Metrics

- Reduction in WISMO Tickets
 - Failed Delivery Reduction
 - Re-delivery Cost Savings
 - Customer Retention
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Risks

Prediction Accuracy

Risk:

Incorrect ETAs reduce customer trust.

Mitigation:

Provide time windows instead of exact times.

Logistics Variability

Risk:

Traffic, COD, and customer availability impact predictions.

Mitigation:

Continuously update ETAs.

Adoption Risk

Risk:

Customers may perceive ETA prediction as a feature rather than a product.

Mitigation:

Demonstrate measurable reductions in support costs and failed deliveries.

Conclusion

The Delivery ETA Intelligence Platform leverages existing logistics infrastructure to solve a major customer pain point: delivery uncertainty.

Rather than rebuilding logistics operations, the platform converts operational data into customer intelligence.

Expected outcomes:

- Reduced customer anxiety
- Improved delivery success rates
- Lower support costs
- Enhanced delivery experience
- New SaaS revenue opportunities

This solution represents a scalable, asset-light opportunity to improve last-mile delivery experiences across India's rapidly growing e-commerce ecosystem.